

SLIDE 1 OF 12

DIRE DAWA UNIVERSITY • DEPT OF SOFTWARE ENGINEERING



Software Engineering Internship Presentation

Architecting LuckyEthio — Enterprise Digital Raffle & Double-Entry Financial Marketplace Platform

Host: SORARDI PLC **Student:** Enter Student Full Name

ID: DDU/UGR/XXXX/XX

Presentation Date: 12/01/2018 E.C. • Submission Date: 10/01/2018 E.C.

SPEAKER NOTE:

"Good morning honorable academic committee, faculty members, and advisors. Today, I am honored to present my final Software Engineering internship project completed at SORARDI PLC."

Agenda & Table of Contents

1. **Introduction & Host Profile:** SORARDI PLC overview and internship objectives.
2. **Problem Statement:** Traditional lottery bottlenecks in Ethiopia.
3. **Project Objectives & Scope:** Functional goals and boundaries.
4. **System Specifications & Feasibility:** Technical, financial, and legal compliance.
5. **Methodology & Architecture:** Agile Scrum, Turborepo monorepo structure.
6. **Key Technical Deliverables:** Double-entry ledger, atomic concurrency, fair RNG.
7. **Live Project Demonstration:** Walkthrough of end-to-end customer and admin flows.
8. **Conclusion & Recommendations:** Academic and industry takeaways.

SPEAKER NOTE:

"Here is the structure of our presentation, flowing from the underlying problem statement to the architectural implementation and live demo."

Host Company Profile: SORARDI PLC

- **Enterprise Profile:** SORARDI PLC is a specialized Ethiopian technology enterprise providing modern enterprise software, cloud applications, and fintech systems.
- **Project Assignment:** Assigned as a Software Engineering Intern to the Core Backend & Architecture team to build the *LuckyEthio* digital raffle infrastructure.
- **Academic Integration:** Direct practical application of university coursework: *Database Systems*, *Web Engineering*, *Computer Security*, and *Agile Project Management*.

SPEAKER NOTE:

"SORARDI PLC provided an exceptional environment to apply advanced software engineering principles directly onto a commercial fintech platform."

Problem Statement: Traditional Raffle Bottlenecks

- **Opaque Manual Draws:** Physical paper lotteries lack digital transparency and verifiable mathematical integrity.
- **Database Concurrency Overselling:** High-traffic sales cause race conditions where seats/tickets sell beyond available inventory.
- **Escrow Delivery Disputes:** Winner prize disbursements often lack dynamic cryptographic proof of handover.
- **Manual Tax Reconciliation:** Difficulty in calculating and remitting statutory 15% VAT to the Ministry of Revenues (MoR).

SPEAKER NOTE:

"These 4 core bottlenecks in Ethiopia's commercial lottery ecosystem defined the technical requirements for our platform."

Project Objectives (General & Specific)

- **General Objective:** To engineer a high-concurrency multi-vendor digital raffle platform featuring double-entry financial ledger accounting and provably fair cryptographic verification.
- **Specific Objectives:**
 - Implement decoupled purchase orders and atomic seat allocation under database transactions.
 - Integrate unified Ethiopian payment adapters (Chapa, Telebirr, CBE Birr, SantimPay).
 - Build a balanced double-entry chart of accounts with 15% VAT escrow automation.
 - Develop multi-entropy SHA-256 commit-reveal RNG and immutable draw snapshot state locks.
 - Enforce Two-Person consensus governance on draws and seller cashouts.

SPEAKER NOTE:

"Our objectives focused on building a secure, auditable, and regulatory-compliant software engine."

System Specifications & Functional Scope

- **Customer Portal (Web & Mobile):** Self-service raffle browsing, decoupled checkout, and instant dynamic QR claim token.
- **Merchant Portal:** Listing creation, 3-angle photo upload, Fayda ID verification, and delivery QR scanner.
- **Agent POS Kiosk & USSD (*157#):** Field ticket sales and offline feature phone accessibility.
- **Admin Governance Console:** Double-entry ledger, trial balance validation, live draw console, and PDPP data privacy manager.

SPEAKER NOTE:

"The system provides comprehensive multi-channel coverage across smartphones, physical kiosks, and 2G feature phones."

Feasibility Analysis

- **Technical Feasibility:** Built with Next.js 14, TypeScript, Prisma ORM, Node.js Crypto, and Tailwind CSS. 100% type-safe with zero proprietary vendor lock-in.
- **Operational Feasibility:** Bilingual English & Amharic (አማርኛ) support, right-aligned mobile ergonomics, and USSD feature phone interface.
- **Economic Feasibility:** Self-sustaining monetization via 8% platform commissions and automated 15% source-deducted VAT tax compliance.
- **Legal & Regulatory Feasibility:** Adheres to National Lottery Administration Proclamation No. 535/2007 and PDPP Proclamation No. 1321/2024.

SPEAKER NOTE:

"Feasibility was verified across technical scalability, local usability, economic viability, and Ethiopian statutory compliance."

Methodology & Turborepo Monorepo Architecture

- **Agile Scrum Methodology:** 2-week sprint cycles, daily standups, and pull request code reviews.
- **Turborepo Monorepo Structure:**
 - `apps/web`: Customer Portal, POS Kiosk, USSD (*157#) Simulator, Public Verifier (Port 3000).
 - `apps/admin`: Admin Operations, Financial Ledger, Live Draw Console (Port 3001).
 - `packages/database`: Prisma schema (18 models), migrations, seed scripts.
 - `packages/shared`: Ledger, Payments, Fair-RNG, Risk, Two-Person, i18n.

SPEAKER NOTE:

"Our monorepo cleanly separates shared business logic from the user-facing web and admin applications."

Double-Entry Financial Ledger Implementation

- Every ticket sale atomically posts balanced journal entries across standard Chart of Accounts:
- **Debit:** 1010 - CASH - TRANSIT (Asset • 100%)
- **Credit:** 2020 - VAT - PAYABLE (Liability • 15% MoR Tax Escrow)
- **Credit:** 2010 - PRIZE - ESCROW (Liability • 77% Seller Holding)
- **Credit:** 4010 - PLATFORM - REVENUE (Revenue • 8% Platform Fee)
- **Trial Balance Status:** 100% Balanced with zero mathematical variance ($\Delta = 0.00$).

SPEAKER NOTE:

"This guarantees that money in transit, prize escrow, and government taxes are balanced down to the exact cent."

Provably Fair Multi-Entropy RNG & Snapshot Lock

- **Pre-Commitment:** Digital SHA-256 commitment published before selling ticket #1.
- **Immutable Snapshot:** When sales close, the exact ticket universe is locked in DrawSnapshot.
- **Two-Person Consensus:** Draw Operator initiates → Super Admin approves.
- **Deterministic Math:** $\text{Winner} = (\text{BigInt}(\text{SHA256}(\text{v2.0} : \text{RaffleID} : \text{Seed} : \text{Sold} : \text{PublicEntropy} : \text{Algo})) \% \text{Sold}) + 1.$
- **Public Verifier:** Instant 1-click independent proof validator at `/verifier`.

SPEAKER NOTE:

"Neither the operator nor the buyer can manipulate the winning outcome once the pre-commitment is published."

Conclusion & Future Recommendations

- **Conclusion:** Successfully architected and deployed a production-grade digital raffle and financial transaction platform for SORARDI PLC, compiling 57 Next.js routes with zero errors.
- **Recommendations for SORARDI PLC:** Implement cloud Hardware Security Modules (HSM) and dedicated telecom SMPP SMS gateways.
- **Recommendations for DDU:** Introduce enterprise monorepo tools (Turborepo) and financial ledger design into software engineering lab coursework.

SPEAKER NOTE:

"In conclusion, this internship successfully demonstrated the application of advanced software engineering in building resilient financial infrastructure."

Live System Demonstration Walkthrough

1. **Flow 1: Customer Ticket Purchase** → Decoupled purchase order, Telebirr payment simulation, and atomic ticket minting.
2. **Flow 2: Double-Entry Financial Ledger** → Review balanced journal entries, 15% VAT escrow, and trial balance on /financials.
3. **Flow 3: Two-Person Provably Fair Live Draw** → Snapshot freeze, dual-authorization, and public verification at /verifier.
4. **Flow 4: Winner Claim QR & Merchant Scanning** → Dynamic QR token generation and merchant delivery escrow release.
5. **Flow 5: Data Privacy Governance** → PDPP Proclamation No. 1321/2024 compliance portal.

SPEAKER NOTE:

"Thank you for your time and attention. I will now proceed to the live demonstration of the LuckyEthio system."